

DNA extraction- CHAOS v1.1

Step 1

1. Place a small piece of coral in the CHAOS solution.
e.g. Put a small piece of coral tissue (less than 1x1x1cm) into 1.5ml tube and add CHAOS solution.
2. Mix it and leave it for more than 3 days at room temp. (Ideally 5 days)
If the size is bigger, you can add more solution. It is better to add 3-5 times volume solution against the volume of the sample. There is no problem in DNA extraction even you have add more CHAOS solution.
3. Leave it for few days before continue DNA extraction. (You can even keep it for years at room temperature)

CHAOS solution:

	100ml	500ml
4M Guanidine thiocyanate (50g/100ml)	50g	250g
0.5% N-lauroyl sarcosine sodium	0.5g	2.5g
25mM Tris pH8	2.5ml	12.5ml
0.1M 2-mercaptoethanol (0.7ml/100ml)	0.7ml	3.5ml
ddH ₂ O	Up to 100ml	Up to 500ml

- Mix it (app. 30min-1 hour).

Step 2

Phenol/chloroform extraction.

1. Add 1X solution, 1X PEB, 2X Phenol/Chloroform and RNaseA (a little- 4µl, or 10mg/ml) in 2.0ml tube.
Eq. 200µl of solution (only transfer the solution, sometimes it will contain some undissolved tissue and mucus but it is ok, just do not get much of them.)
200µl (equal vol. of solution) of PEB* (see below for protocol)
400µl (equal vol. of solution & PEB) of Phenol/Chloroform
2. Mix well (turn over the tube repeatedly) then centrifuge it 14000rpm for 5 min.
3. Move upper layer to new tube then add 2X Phenol/Chloroform.
4. Mix well (same as above) then centrifuge it 14000rpm for 5 min.
5. Move upper layer (not all but leave some) to new 1.5ml tube.
6. Add 2X (equal vol of Phenol/Chloroform) of Isopropanol or 400µl (2X) of 99.8% ethanol.
7. Mix gently (just turn over repeatedly).
8. The DNA will visually appear, then centrifuge it 14000rpm for 10 min.
If DNA cannot be seen visually, centrifuge it 14000rpm for 20 min.
9. Then discard the isopropanol or ethanol, and add about 200-800 µl of Isopropanol or ethanol for rinse.
10. And microcentrifuge if DNA is few.
11. Discard isopropanol or ethanol and air dry.
12. Add 50µl of pre-heated TE(65 °C) .
13. Mix and store the DNA in -20 °C.

PEB (phenol extraction buffer)

	100 ml	500ml
100mM TrisCl-pH8	0.01ml (1M TrisCl)	
10mM EDTA		
0.1% SDS		